

Exercise 18.4

1. Find, from the definition, the derivative of

a) e^{2x}

→ let $f(x) = e^{2x}$

$$f(x+h) = e^{2(x+h)}$$

Where h is small change in x .

From the definition of derivative,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^{2(x+h)} - e^{2x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^{2x+2h} - e^{2x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^{2x} \cdot e^{2h} - e^{2x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^{2x} (e^{2h} - 1)}{h}$$

$$= \lim_{h \rightarrow 0} e^{2x} \times \left(\frac{e^{2h} - 1}{2h} \right) \times 2$$

$$= e^{2x} \times 1 \times 2$$

$$= 2e^{2x}$$

b) e^{bx+c}

→ let $f(x) = e^{bx+c} = e^{bx} \cdot e^c$

$$f(x+h) = e^{b(x+h)+c} = e^{bx} \cdot e^{bh} \cdot e^c$$

Where h is small change in x .

From the definition of derivative,

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{e^{bx+c} \cdot e^{bh} \cdot e^c - e^{bx} \cdot e^c}{h} \\
 &= \lim_{h \rightarrow 0} \frac{e^{bx+c} (e^{bh} - 1)}{h} \\
 &= \lim_{h \rightarrow 0} e^{bx+c} \left(\frac{e^{bh} - 1}{bh} \right) \times b \\
 &= e^{bx+c} \times 1 \times b \\
 &= b e^{bx+c}
 \end{aligned}$$

c) $\ln(ax+b)$

→ let $f(x) = \ln(ax+b)$

$$\begin{aligned}
 f(x+h) &= \ln(a(x+h)+b) \\
 &= \ln(ax+ah+b)
 \end{aligned}$$

Where h is small change in x .

From the definition of derivative

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\ln(ax+ah+b) - \ln(ax+b)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\ln \left(\frac{ax+ah+b}{ax+b} \right)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\ln \left(1 + \frac{ah}{ax+b} \right)}{\frac{ah}{ax+b}} \times \frac{a}{ax+b} \\
 &= 1 \times \frac{a}{ax+b} = \frac{a}{ax+b}
 \end{aligned}$$

d) $\ln x^2$

→ let $f(x) = \ln x^2 = 2 \ln x$

$f(x+h) = \ln (x+h)^2 = 2 \ln (x+h)$

Where h is small change in x .

By definition of derivative,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \ln(x+h) - 2 \ln x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \left[\ln \left(\frac{x+h}{x} \right) \right]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \left[\ln \left(1 + \frac{h}{x} \right) \right]}{\frac{h}{x}} \times \frac{1}{x}$$

$$= 2 \times 1 \times \frac{1}{x}$$

$$= \frac{2}{x}$$

e) $\ln \left(\frac{x}{5} \right)$

→ let $f(x) = \ln \left(\frac{x}{5} \right) = \ln x - \ln 5$

$f(x+h) = \ln \left(\frac{x+h}{5} \right)$

Where h is small change in x .

By definition of derivative,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(\frac{x+h}{x}\right) - \ln\left(\frac{x}{x}\right)}{h}$$

$$= \lim_{h \rightarrow 0} \ln \left(\frac{\frac{x+h}{x}}{\frac{x}{x}} \right)$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(1 + \frac{h}{x}\right)}{\frac{h}{x}} \times \frac{1}{x}$$

$$= 1 \times \frac{1}{x}$$

$$= \frac{1}{x}$$

f) $\log_{10} x$

→ let $f(x) = \log_{10} x = \log_{10} e \cdot \log_e x$

$$f(x+h) = \log_{10} e \cdot \log_e (x+h)$$

Where h is small change in x .

By definition of derivative,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\log_{10} e \cdot \log_e (x+h) - \log_{10} e \cdot \log_e x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\log_{10} e [\log_e (x+h) - \log_e x]}{h}$$

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$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \log_{10} e \left[\log_e \left(\frac{x+h}{x} \right) \right] \times \frac{1}{h} \\
 &= \lim_{h \rightarrow 0} \log_{10} e \left[\ln \left(\frac{x+h}{x} \right) \right] \quad [\because \log_e x = \ln x] \\
 &= \lim_{h \rightarrow 0} \frac{\log_{10} e \left[\ln \left(1 + \frac{h}{x} \right) \right]}{\frac{h}{x}} \times \frac{1}{x} \\
 &= \log_{10} e \times 1 \times \frac{1}{x} \\
 &= \frac{1}{x} \times \log_{10} e
 \end{aligned}$$

2. Find, the derivative of

a) $\ln(\cos x)$

→ let, $y = \ln(\cos x)$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d \ln(\cos x)}{dx}$$

$$= \frac{d \ln(\cos x)}{d \cos x} \times \frac{d \cos x}{dx}$$

$$= \frac{1}{\cos x} \times -\sin x$$

$$= -\frac{\sin x}{\cos x}$$

$$= -\tan x$$

b) $\ln(\ln x)$

→ let $y = \ln(\ln x)$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d \ln(\ln x)}{dx}$$

$$= \frac{d \ln(\ln x)}{d(\ln x)} \times \frac{d(\ln x)}{dx}$$

$$= \frac{1}{\ln x} \times \frac{1}{x}$$

$$= \frac{1}{x \ln x} \quad \#$$

c) $\ln(x + \cot x)$

→ let $y = \ln(x + \cot x)$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d \ln(x + \cot x)}{dx}$$

$$= \frac{d \ln(x + \cot x)}{d(x + \cot x)} \times \frac{d(x + \cot x)}{dx}$$

$$= \frac{1}{x + \cot x} \times [1 + (-\operatorname{cosec}^2 x)]$$

$$= \frac{1 - \operatorname{cosec}^2 x}{x + \cot x}$$

$$= \frac{-\cot^2 x}{x + \cot x} \quad \#$$

d) $\ln(1+x+x^2)$

→ let $y = \ln(1+x+x^2)$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d \ln(1+x+x^2)}{dx}$$

$$\begin{aligned}
 &= \frac{d \ln(1+x+x^2)}{d(1+x+x^2)} \times \frac{d(1+x+x^2)}{dx} \\
 &= \frac{1}{1+x+x^2} \times (1+2x) \\
 &= \frac{1+2x}{1+x+x^2} \quad \checkmark
 \end{aligned}$$

e) $e^{\cos x}$

→ let $y = e^{\cos x}$

differentiating with respect to x .

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d e^{\cos x}}{d \cos x} \times \frac{d \cos x}{dx} \\
 &= e^{\cos x} \times (-\sin x) \\
 &= -\sin x e^{\cos x} \quad \checkmark
 \end{aligned}$$

f) e^{1+x+x^2}

→ let $y = e^{1+x+x^2}$

differentiating with respect to x .

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d e^{1+x+x^2}}{d(1+x+x^2)} \times \frac{d(1+x+x^2)}{dx} \\
 &= e^{1+x+x^2} \times (1+2x) \\
 &= e^{1+x+x^2} \cdot (1+2x) \quad \checkmark
 \end{aligned}$$

g) $e^{\sin(\ln x)}$

→ let $y = e^{\sin(\ln x)}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d e^{\sin(\ln x)}}{d \sin(\ln x)}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d e^{\sin(\ln x)}}{d \sin(\ln x)} \times \frac{d \sin(\ln x)}{d (\ln x)} \times \frac{d (\ln x)}{dx} \\ &= e^{\sin(\ln x)} \times \cos(\ln x) \times \frac{1}{x} \\ &= \frac{1}{x} \cos(\ln x) \times e^{\sin(\ln x)}\end{aligned}$$

#.

h) $\ln(e^{nx} + e^{-nx})$

→ let $y = \ln(e^{nx} + e^{-nx})$

differentiating with respect to x .

$$\begin{aligned}\frac{dy}{dx} &= \frac{d [\ln(e^{nx} + e^{-nx})]}{dx} \\ &= \frac{d \ln(e^{nx} + e^{-nx})}{d (e^{nx} + e^{-nx})} \times \frac{d (e^{nx} + e^{-nx})}{dx} \\ &= \frac{1}{e^{nx} + e^{-nx}} \times \left[\frac{d e^{nx}}{dx} + \frac{d e^{-nx}}{dx} \right] \\ &= \frac{1}{e^{nx} + e^{-nx}} \left[\frac{d e^{nx}}{dx} \times \frac{dx}{dx} + \frac{d e^{-nx}}{d(-nx)} \times \frac{d(-nx)}{dx} \right] \\ &= \frac{1}{e^{nx} + e^{-nx}} [e^{nx} \times n + e^{-nx} \times -n] \\ &= \frac{1}{e^{nx} + e^{-nx}} [n e^{nx} - n e^{-nx}] \\ &= \frac{n}{e^{nx} + e^{-nx}} [e^{nx} - e^{-nx}]\end{aligned}$$

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3. Find the derivative of.

a) $x^5 e^{ax}$

→ let $y = x^5 \cdot e^{ax}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d}{dx} [x^5 \cdot e^{ax}]$$

$$= x^5 \frac{d e^{ax}}{dx} + e^{ax} \cdot \frac{d x^5}{dx}$$

$$= x^5 \cdot \frac{d e^{ax}}{d ax} \times \frac{d ax}{dx} + e^{ax} \cdot 5x^4$$

$$= x^5 \cdot e^{ax} \times a + e^{ax} \cdot 5x^4$$

$$= e^{ax} [ax^5 + 5x^4] \quad \checkmark$$

b) $e^{ax} \cdot \cos bx$

→ let $y = e^{ax} \cdot \cos bx$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d}{dx} [e^{ax} \cdot \cos bx]$$

$$= \frac{d e^{ax}}{dx} \times \cos bx + e^{ax} \times \frac{d \cos bx}{dx}$$

$$= \cos bx \times \frac{d e^{ax}}{d ax} \times \frac{d ax}{dx} + e^{ax} \times \frac{d \cos bx}{d bx} \times \frac{d bx}{dx}$$

$$= \cos bx \times e^{ax} \times a + e^{ax} \times (-\sin bx) \times b$$

$$= a \cos bx \cdot e^{ax} - b \sin bx e^{ax}$$

$$= e^{ax} (a \cos bx - b \sin bx) \quad \checkmark$$

c) $(\sin x + \cos x) e^{ax}$

→ let $y = (\sin x + \cos x) \cdot e^{ax}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d[(\sin x + \cos x) \cdot e^{ax}]}{dx}$$

$$= (\sin x + \cos x) \frac{de^{ax}}{dx} + e^{ax} \frac{d(\sin x + \cos x)}{dx}$$

$$= (\sin x + \cos x) \times \frac{de^{ax}}{dax} \times \frac{dax}{dx} + e^{ax} \left[\frac{d\sin x}{dx} + \frac{d\cos x}{dx} \right]$$

$$= (\sin x + \cos x) \times e^{ax} \times a + e^{ax} [\cos x + (-\sin x)]$$

$$= e^{ax} [a\sin x + a\cos x + \cos x - \sin x]$$

$$= e^{ax} [(a+1)\cos x + (a-1)\sin x]$$

d) $e^{ax}(a\sin bx - b\cos bx)$

→ let $y = e^{ax}(a\sin bx - b\cos bx)$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d[e^{ax}(a\sin bx - b\cos bx)]}{dx}$$

$$= e^{ax} \frac{d(a\sin bx - b\cos bx)}{dx} + (a\sin bx - b\cos bx) \frac{de^{ax}}{dx}$$

$$= e^{ax} \left[\frac{da\sin bx}{dbx} \times \frac{dbx}{dx} - \frac{db\cos bx}{dbx} \times \frac{dbx}{dx} \right] +$$

$$(a\sin bx - b\cos bx) \frac{de^{ax}}{dax} \times \frac{dax}{dx}$$

$$= e^{ax} [a\cos bx \times b + b\sin bx \times b] + (a\sin bx - b\cos bx) \times e^{ax} \times a$$

$$= e^{ax} [ab\cos bx + b^2\sin bx] + a \cdot e^{ax}(a\sin bx - b\cos bx)$$

$$= e^{ax} [ab\cos bx + b^2\sin bx + a^2\sin bx - ab\cos bx]$$

$$= e^{ax} [(a^2 + b^2)\sin bx]$$

$$= (a^2 + b^2) e^{ax} \cdot \sin bx$$

e) $x^2 \ln(1+x)$

→ let $y = x^2 \ln(1+x)$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d[x^2 \ln(1+x)]}{dx}$$

$$= x^2 \frac{d \ln(1+x)}{dx} + \ln(1+x) \cdot \frac{dx^2}{dx}$$

$$= x^2 \frac{d(\ln(1+x))}{d(1+x)} \times \frac{d(1+x)}{dx} + \ln(1+x) \cdot 2x$$

$$= x^2 \times \frac{1}{(1+x)} \times 1 + 2x \ln(1+x)$$

$$= \frac{x^2}{1+x} + 2x \ln(1+x)$$

$$= x \left[\frac{x}{1+x} + 2 \ln(1+x) \right]$$

f) $e^{5x} \cdot \sin(\ln x)$

→ let $y = e^{5x} \cdot \sin(\ln x)$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d[e^{5x} \sin(\ln x)]}{dx}$$

$$= e^{5x} \frac{d \sin(\ln x)}{dx} + \sin(\ln x) \cdot \frac{de^{5x}}{dx}$$

$$= e^{5x} \times \frac{d \sin(\ln x)}{d(\ln x)} \times \frac{d \ln x}{dx} + \sin(\ln x) \times \frac{de^{5x}}{d5x} \times \frac{d5x}{dx}$$

$$= e^{5x} \times \cos(\ln x) \times \frac{1}{x} + \sin(\ln x) \times e^{5x} \times 5$$

$$= e^{5x} \left[\frac{\cos(\ln x)}{x} + 5 \sin(\ln x) \right]$$

g) $\frac{e^x}{\ln x}$

→ let $y = \frac{e^x}{\ln x}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d \left[\frac{e^x}{\ln x} \right]}{dx}$$

$$= \frac{\ln x \frac{de^x}{dx} - e^x \frac{d(\ln x)}{dx}}{(\ln x)^2}$$

$$= \frac{\ln x \times e^x - e^x \times \frac{1}{x}}{(\ln x)^2}$$

$$= \frac{e^x [x \ln x - 1]}{x (\ln x)^2}$$

h) $\frac{\ln x}{1 + \ln x}$

→ let $y = \frac{\ln x}{1 + \ln x}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d \left[\frac{\ln x}{1 + \ln x} \right]}{dx}$$

$$= \frac{(1 + \ln x) \frac{d \ln x}{dx} - \ln x \frac{d(1 + \ln x)}{dx}}{(1 + \ln x)^2}$$

$$= \frac{(1 + \ln x) \times \frac{1}{x} - \ln x \left[\frac{1}{x} \right]}{(1 + \ln x)^2}$$

$$(1 + \ln x)^2$$

$$= \frac{\frac{1}{x} [1 + \ln x - \ln x]}{(1 + \ln x)^2}$$

$$= \frac{1}{x(1 + \ln x)^2} \quad \text{Ans.}$$

i) $\frac{\ln x}{a^2 + x^2}$

→ let $y = \frac{\ln x}{a^2 + x^2}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{\ln x}{a^2 + x^2} \right)$$

$$= \frac{(a^2 + x^2) \frac{d}{dx} \ln x - \ln x \frac{d}{dx} (a^2 + x^2)}{(a^2 + x^2)^2}$$

$$= \frac{(a^2 + x^2) \times \frac{1}{x} - \ln x \times 2x}{(a^2 + x^2)^2}$$

$$= \frac{a^2 + x^2 - 2x^2 \ln x}{x(a^2 + x^2)^2} \quad \text{Ans.}$$

j) $\frac{e^{ax}}{\sin bx}$

→ let $y = \frac{e^{ax}}{\sin bx}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{e^{ax}}{\sin bx} \right)$$

$$= \frac{\sin bx \cdot \frac{d}{dx} e^{ax} - e^{ax} \cdot \frac{d}{dx} \sin bx}{(\sin bx)^2}$$

$$= \frac{\sin bx \frac{d e^{ax}}{dx} - e^{ax} \frac{d \sin bx}{dx}}{(\sin bx)^2}$$

$$= \frac{1}{\sin^2 bx} \left[\sin bx \times \frac{d e^{ax}}{dx} \times \frac{d ax}{dx} - e^{ax} \times \frac{d \sin bx}{dx} \times \frac{d bx}{dx} \right]$$

$$= \frac{e^{ax} (a \sin bx - b \cos bx)}{\sin^2 bx} \quad \therefore$$

4. Find, $\frac{dy}{dx}$ when,

a) $\sin y + \ln y = x^2 + 18x + 3$

differentiating with respect to x .

$$\frac{d \sin y}{dx} + \frac{d \ln y}{dx} = \frac{d(x^2 + 18x + 3)}{dx}$$

$$\text{or, } \frac{d \sin y}{dy} \times \frac{dy}{dx} + \frac{d \ln y}{dy} \times \frac{dy}{dx} = 2x + 18$$

$$\text{or, } \cos y \times \frac{dy}{dx} + \frac{1}{y} \times \frac{dy}{dx} = 2x + 18$$

$$\text{or, } \frac{dy}{dx} \left[\cos y + \frac{1}{y} \right] = 2x + 18$$

$$\text{or, } \frac{dy}{dx} \left[\frac{y \cos y + 1}{y} \right] = 2x + 18$$

$$\text{or, } \frac{dy}{dx} = \frac{2y(x+9)}{1+y \cos y}$$

$$\therefore \frac{dy}{dx} = \frac{2y(x+9)}{(1+y \cos y)} \quad \therefore$$

b) $x^2 + y^2 = \ln(xy)$

→ differentiating with respect to x .

$$\frac{d x^2}{d x} + \frac{d y^2}{d x} = \frac{d \ln(xy)}{d x}$$

$$\text{or, } 2x + \frac{dy^2}{dy} \times \frac{dy}{dx} = \frac{d \ln(xy)}{d(xy)} \times \frac{d(xy)}{dx}$$

$$\text{or, } 2x + 2y \times \frac{dy}{dx} = \frac{1}{(xy)} \times \left(1 + \frac{dy}{dx} \right)$$

$$\text{or, } 2x + 2y \times \frac{dy}{dx} = \frac{1}{xy} + \frac{1}{xy} \times \frac{dy}{dx}$$

$$\text{or, } \frac{dy}{dx} \left[2y - \frac{1}{xy} \right] = \frac{1}{xy} - 2x$$

$$\text{or, } \frac{dy}{dx} \left[\frac{2xy + 2y^2 - 1}{xy} \right] = \frac{1 - 2x^2 - 2xy}{xy}$$

$$\text{or, } \frac{dy}{dx} = \frac{1 - 2x^2 - 2xy}{2xy + 2y^2 - 1}$$

$$\therefore \frac{dy}{dx} = \frac{1 - 2x^2 - 2xy}{2xy + 2y^2 - 1}$$

c) $x = y \ln xy$

→ differentiating with respect to x .

$$\frac{dx}{dx} = \frac{d[y \ln xy]}{dx}$$

$$\text{or, } 1 = y \frac{d \ln xy}{dx} + \ln xy \frac{dy}{dx}$$

$$\text{or, } 1 = y \times \frac{d \ln xy}{d xy} \times \frac{d xy}{dx} + \ln xy \times \frac{dy}{dx}$$

$$\text{or, } 1 = y \times \frac{1}{xy} \times \left[y + x \times \frac{dy}{dx} \right] + \ln xy \times \frac{dy}{dx}$$

$$\text{or, } 1 = \frac{y}{x} + \frac{x}{x} \times \frac{dy}{dx} + \ln xy \times \frac{dy}{dx}$$

$$\text{or, } 1 = \frac{y}{x} + \frac{dy}{dx} + \ln xy \times \frac{dy}{dx}$$

$$\text{or, } 1 = \frac{y}{x} + (1 + \ln xy) \times \frac{dy}{dx}$$

$$\text{or, } \frac{dy}{dx} = \left(1 - \frac{y}{x}\right) \times \frac{1}{(1 + \ln xy)}$$

$$\text{or, } \frac{dy}{dx} = \frac{x - y}{x(1 + \ln xy)}$$

$$\therefore \frac{dy}{dx} = \frac{x - y}{x(1 + \ln xy)}$$

$$d) e^{xy} = xy$$

→ differentiating with respect to x .

$$\frac{d e^{xy}}{dx} = \frac{d xy}{dx}$$

$$\text{or, } \frac{d e^{xy}}{dx} \times \frac{d xy}{dx} = \frac{d xy}{dx}$$

$$\text{or, } e^{xy} \times \left[x \frac{dy}{dx} + y \right] = \left[x \frac{dy}{dx} + y \right]$$

$$\text{or, } e^{xy} \times x \frac{dy}{dx} + e^{xy} \times y = x \frac{dy}{dx} + y$$

$$\text{or, } e^{xy} \times x \frac{dy}{dx} - x \frac{dy}{dx} = y - e^{xy} \times y$$

$$\text{or, } x \frac{dy}{dx} [e^{xy} - 1] = y [1 - e^{xy}]$$

$$\text{or, } x \frac{dy}{dx} [e^{xy} - 1] = -y [e^{xy} - 1]$$

$$\text{or, } \frac{dy}{dx} = -\frac{y}{x}$$

$$\therefore \frac{dy}{dx} = -\frac{y}{x}$$

e) $xy = \ln(x^2 + y^2)$

→ differentiating with respect to x .

$$\frac{dxy}{dx} = \frac{d \ln(x^2 + y^2)}{dx}$$

$$\text{or, } x \times \frac{dy}{dx} + y \times \frac{dx}{dx} = \frac{d \ln(x^2 + y^2)}{d(x^2 + y^2)} \times \frac{d(x^2 + y^2)}{dx}$$

$$\text{or, } x \times \frac{dy}{dx} + y = \frac{1}{x^2 + y^2} \times \left[\frac{dx^2}{dx} + \frac{dy^2}{dy} \times \frac{dy}{dx} \right]$$

$$\text{or, } x \times \frac{dy}{dx} + y = \frac{1}{x^2 + y^2} \left[2x + 2y \times \frac{dy}{dx} \right]$$

$$\text{or, } x \times \frac{dy}{dx} + y = \frac{2x}{x^2 + y^2} + \frac{2y}{x^2 + y^2} \times \frac{dy}{dx}$$

$$\text{or, } x \times \frac{dy}{dx} - \frac{2y}{x^2 + y^2} \times \frac{dy}{dx} = \frac{2x}{x^2 + y^2} - y$$

$$\text{or, } \frac{dy}{dx} \left[x - \frac{2y}{x^2 + y^2} \right] = \frac{2x - y(x^2 + y^2)}{x^2 + y^2}$$

$$\text{or, } \frac{dy}{dx} \left(\frac{x(x^2 + y^2) - 2y}{x^2 + y^2} \right) = \frac{2x - y(x^2 + y^2)}{x^2 + y^2}$$

$$\text{or, } \frac{dy}{dx} = \frac{2x - y(x^2 + y^2)}{x(x^2 + y^2) - 2y}$$

$$\therefore \frac{dy}{dx} = \frac{2x - y(x^2 + y^2)}{x(x^2 + y^2) - 2y}$$

f) $x^3 y^4 = (x+y)^7$

→ Taking $\ln$ on both side,

$$\ln x^3 y^4 = \ln (x+y)^7$$

$$\text{or, } \ln x^3 + \ln y^4 = 7 \ln (x+y)$$

differentiating with respect to x .

$$\frac{d(\ln x^3)}{dx} + \frac{d(\ln y^4)}{dx} = 7 \frac{d \ln (x+y)}{dx}$$

$$\frac{d}{dx} 3 \ln x + \frac{d}{dy} 4 \ln y = 7 \frac{d \ln(x+y)}{d(x+y)} \times \frac{d(x+y)}{dx}$$

$$\text{or, } 3 \times \frac{1}{x} + 4 \times \frac{d \ln y}{dy} \times \frac{dy}{dx} = 7 \times \frac{1}{x+y} \times \left(1 + \frac{dy}{dx}\right)$$

$$\text{or, } \frac{3}{x} + \frac{4}{y} \times \frac{dy}{dx} = \frac{7}{x+y} + \frac{7}{x+y} \times \frac{dy}{dx}$$

$$\text{or, } \frac{4}{y} \times \frac{dy}{dx} - \frac{7}{x+y} \times \frac{dy}{dx} = \frac{7}{x+y} - \frac{3}{x}$$

$$\text{or, } \frac{dy}{dx} \left[\frac{4}{y} - \frac{7}{x+y} \right] = \frac{7x - 3x - 3y}{x(x+y)}$$

$$\text{or, } \frac{dy}{dx} \left[\frac{4x + 4y - 7y}{y(x+y)} \right] = \frac{4x - 3y}{x(x+y)}$$

$$\text{or, } \frac{dy}{dx} \times \left[\frac{4x - 3y}{y} \right] = \frac{4x - 3y}{x}$$

$$\text{or, } \frac{dy}{dx} = \frac{y}{x}$$

$$\therefore \frac{dy}{dx} = \frac{y}{x}$$

5.

a) Find, $\frac{dy}{dx}$ when

$$\text{i) } x = e^\theta \cdot \cos \theta, \quad y = e^\theta \cdot \sin \theta$$

$$\rightarrow x = e^\theta \cdot \cos \theta \rightarrow \text{(i)}$$

$$y = e^\theta \cdot \sin \theta \rightarrow \text{(ii)}$$

Taking ~~ln~~ on both side,~~ln~~ differentiating (i) with respect to θ .

$$\frac{dx}{d\theta} = \frac{d[e^\theta \cdot \cos \theta]}{d\theta}$$

$$\begin{aligned}
 &= e^{\theta} \cdot \frac{d \cos \theta}{d \theta} + \cos \theta \cdot \frac{d e^{\theta}}{d \theta} \\
 &= e^{\theta} \cdot (-\sin \theta) + \cos \theta \cdot e^{\theta} \\
 &= e^{\theta} (-\sin \theta + \cos \theta) \\
 \frac{d m}{d \theta} &= e^{\theta} (\cos \theta - \sin \theta) \rightarrow \text{(iii)}
 \end{aligned}$$

differentiating (ii) with respect to θ

$$\begin{aligned}
 \frac{d y}{d \theta} &= \frac{d (e^{\theta} \cdot \sin \theta)}{d \theta} \\
 &= e^{\theta} \cdot \frac{d \sin \theta}{d \theta} + \sin \theta \cdot \frac{d e^{\theta}}{d \theta} \\
 &= e^{\theta} \cdot \cos \theta + \sin \theta \cdot e^{\theta} \\
 &= e^{\theta} (\sin \theta + \cos \theta) \rightarrow \text{(iv)}
 \end{aligned}$$

Now,

$$\begin{aligned}
 \frac{d y}{d n} &= \frac{d y}{d \theta} \times \frac{d \theta}{d n} \\
 &= \frac{e^{\theta} (\sin \theta + \cos \theta) \times 1}{e^{\theta} (\cos \theta - \sin \theta)} \\
 &= \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}
 \end{aligned}$$

ii) $y = \ln t + \sin t$, $n = e^t + \cos t$

$\rightarrow y = \ln t + \sin t \rightarrow \text{(i)}$

$n = e^t + \cos t \rightarrow \text{(ii)}$

differentiating (i) with respect to t .

$$\frac{d y}{d t} = \frac{d \ln t}{d t} + \frac{d \sin t}{d t}$$

$$\frac{dy}{dt} = \frac{1}{t} + \cos t \rightarrow$$

differentiating (i) with respect to t .

$$\begin{aligned}\frac{d\ln}{dt} &= \frac{d e^t}{dt} + \frac{d \cos t}{dt} \\ &= e^t + (-\sin t) \\ &= e^t - \sin t\end{aligned}$$

$$\begin{aligned}\frac{dy}{d\ln} &= \frac{dy}{dt} \times \frac{dt}{d\ln} \\ &= \left(\frac{1}{t} + \cos t \right) \times \frac{1}{e^t - \sin t} \\ &= \frac{1 + t \cos t}{t(e^t - \sin t)}\end{aligned}$$

b) If $y = e^{\sin 2t}$ and $x = e^{\cos 2t}$, then show that

$$\frac{dy}{dx} = \frac{-y \ln x}{x \ln y}$$

$$\rightarrow y = e^{\sin 2t} \rightarrow (i)$$

Taking $\ln$ on both side,

$$\ln y = \ln e^{\sin 2t}$$

$$\ln y = \sin 2t \ln e$$

$$\ln y = \sin 2t \rightarrow (ii)$$

differentiating with respect to t .

$$\frac{d \ln y}{dt} = \frac{d \sin 2t}{dt}$$

$$\frac{d \ln y}{dy} \times \frac{dy}{dt} = \frac{d \sin 2t}{d 2t} \times \frac{d 2t}{dt}$$

$$\frac{1}{y} \times \frac{dy}{dt} = \cos 2t \times 2$$

$$\frac{dy}{dt} = 2y \cos 2t$$

$$x = e^{\cos 2t} \rightarrow \text{(iii)}$$

Taking $\ln$ on both side,

$$\ln x = \ln e^{\cos 2t}$$

$$\ln x = \cos 2t \ln e$$

$$\ln x = \cos 2t \rightarrow \text{(iv)}$$

differentiating with respect to t .

$$\frac{d \ln x}{dt} = \frac{d \cos 2t}{dt}$$

$$\text{or, } \frac{d \ln x}{dx} \times \frac{dx}{dt} = \frac{d \cos 2t}{d 2t} \times \frac{d 2t}{dt}$$

$$\text{or, } \frac{1}{x} \times \frac{dx}{dt} = -\sin 2t \times 2$$

$$\text{or, } \frac{dx}{dt} = -2x \sin 2t$$

Using chain rule,

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= 2y \cos 2t \times \frac{1}{-2x \sin 2t}$$

$$= \frac{y \ln x}{-x \ln y}$$

$$\therefore \frac{dy}{dx} = -\frac{y \ln x}{x \ln y}$$

Proved

6. Differentiate

a) e^{2x} with respect to e^x

$$\begin{aligned} & \frac{d e^{2x}}{d e^x} \\ &= \frac{d e^{2x}}{dx} \times \frac{dx}{d e^x} \\ &= \frac{d e^{2x}}{dx} \\ &= e^{2x} \times 2 \\ &= 2e^{2x} \end{aligned}$$

b) $\ln \sec x$ w.r.t. $\tan x$

$$\begin{aligned} & \frac{d \ln \sec x}{d \tan x} \\ &= \frac{d \ln \sec x}{d \sec x} \times \frac{d \sec x}{d \tan x} \\ &= \frac{1}{\sec x} \times \sec x \cdot \tan x \\ &= \tan x \end{aligned}$$